

A nonparametric copula approach to predict heart rate trajectories from single overnight wearable accelerometry

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Abstract— Wearable devices provide continuous, non-invasive monitoring of heart rate and motion at home, offering significant potential to advance precision health and medicine. Conventional heart rate prediction methods often depend on physiological signals through proprietary algorithms or rigid parametric forms. In this study, we explore a copula-based transformation method with a local log-quadratic likelihood kernel estimator and a nearest-neighbor bandwidth (Tll2nn) to estimate heart rate trajectories from only overnight triaxial accelerometer data. We test this approach on Apple Watch data from ~30 healthy participants during sleep. Our model effectively captures nonlinear dependencies between motion-derived features and heart rate fluctuations, outperforming a baseline prediction of constant mean heart rate in 20 of 26 subjects. We conclude that our methodology enables a novel, interpretable, computationally efficient and at-scale automation of heart rate trajectory tracking without relying on a physiological signal, even using data from a single overnight.

Clinical Relevance— Automated tracking of heart rate through widely-utilized wearable devices can revolutionize remote patient monitoring. The technology enables early detection of physiological irregularities and improved management of sleep disorders and cardiovascular diseases, supporting timely intervention and enhanced health outcomes.

I. INTRODUCTION

Accurate heart rate monitoring is vital for personalized health, supporting the detection of cardiac irregularities, optimization of physical activity, and assessment of stress. However, obtaining high-quality longitudinal heart rate (HR) data is costly and typically requires specialized sensors like photoplethysmography (PPG) or electrocardiograms (ECG) [1]. In contrast, accelerometers—often built into consumer-grade wearables—offer an accessible alternative for tracking physical activity. There is growing interest in leveraging accelerometer data to estimate heart rate using stochastic dependence analysis [3], especially during periods of minimal movement such as sleep; in these low-activity conditions, accelerometers can capture subtle vibrations and ballistocardiographic signals from the cardiovascular system [6].

Existing motion-based HR prediction algorithms leverage complex machine learning techniques like Random Forests and Support Vector Machines (SVMs) or rely on CNNs to combine motion with HR measurements [2]. While effective, these models require significant compute, and are unable to capture intricate, non-linear relationships between heart rate and motion data in certain contexts.

Copula-based models offer a more flexible and robust

framework for modeling dependencies. A copula function links the marginal distributions of random variables to their joint distribution, separating dependency structures from individual behaviors [3]. Though copulas have been scarcely used in this context, we hypothesize that this decomposition is advantageous for modeling multivariate time-series data.

A bivariate copula function C is the joint cumulative distribution function of a continuous random vector (X_1, X_2) whose marginals are uniform over $[0, 1]$ [3]. Let F_j be the marginal cumulative distribution function (CDF) of X_j for $j = 1, 2$ and F be the joint CDF. We apply the probability integral transform and define uniform random variable $U_j = F_j(X_j)$. Then, the cumulative distribution function (CDF) of (U_1, U_2) is the copula of (X_1, X_2) , denoted by C (Equation 1).

$$C(u_1, u_2) = P(U_1 \leq u_1, U_2 \leq u_2) \quad (1)$$

The joint distribution can be decomposed into two components: the copula and marginals by Sklar's Theorem (Equation 2) [4]. This is illustrated schematically in Fig. 1.

$$\begin{aligned} F(x_1, x_2) &= P(X_1 \leq x_1, X_2 \leq x_2) \\ &= P(U_1 \leq F_1(x_1), U_2 \leq F_2(x_2)) \\ &= C(F_1(x_1), F_2(x_2)) \end{aligned} \quad (2)$$

Archimedean (Clayton, Frank etc.) copulas and Elliptical (Gaussian or t-copulas) are well-established parametric forms that have been used to capture dependencies between short-term heart rate and accelerometer-derived

motion data [5]. However, their rigid assumptions and risk for boundary bias—particularly when kernel density estimators (KDEs) are employed—limit their effectiveness in modeling tail dependencies. KDEs underestimate density near data

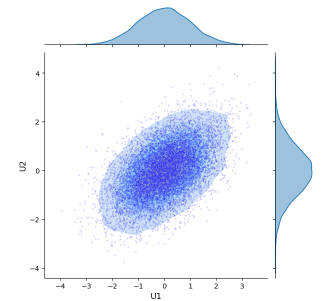


Fig 1. Contour of a synthetic Gaussian multivariate distribution with marginals on the top and right.

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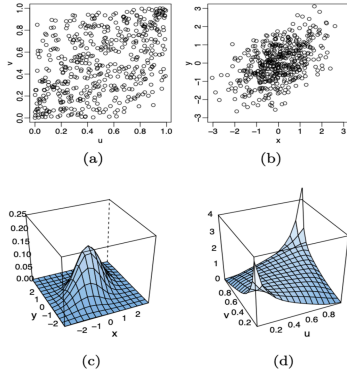


Fig 2: The transformation estimator: (a) copula sample (b) transformed sample (c) kernel density estimate for transformed sample (d) kernel estimate of the copula density. Adapted from Nagler, T. (2018). *Kernel Smoothing for Bivariate Copula Densities*

boundaries, leading to inaccuracies in copula estimation [4]. To overcome this limitation, we leverage transformation-based and local-likelihood density estimation techniques. The transformation method maps data onto $\mathbb{R}^2$ to enable standard KDE application within the transformed domain. Meanwhile, local-likelihood density estimation uses the probit function to transform uniform marginals into a normal distribution. Density estimation via local-likelihood methods and k-Nearest-Neighbor (kNN) bandwidth selection further enhance smoothness [4], integrating the simplicity of parametric models with the flexibility of nonparametric estimators.

In this study, we explore the TIl2nn technique to fit individual nonparametric copulas to HR and accelerometer data from 26 healthy subjects during sleep. We demonstrate that only 3 to 8 hours of overnight triaxial acceleration data, combined with limited heart rate measurements, can reliably estimate full heart rate trajectories during sleep. In the remainder of this paper, we detail our implementation of the copula alongside findings and implications. Our results demonstrate the model's robustness in capturing inter-individual variability, especially when direct skin contact for continuous HR monitoring is impractical.

II. METHODS

A. Dataset and Cohort Selection

We used the publicly available SleepAccel dataset from the National Sleep Research Resource (NSRR), which contains overnight heart rate and triaxial accelerometer data from an Apple watch and expert-labeled sleep stages for 31 healthy participants [7]. Five subjects' data were excluded due to data quality issues (such as inconsistent sampling variability) that resulted in unreliable HR monitoring. Data from the remaining 26 subjects were used in this study.

B. Data Preprocessing

For all 26 subjects, wake periods were identified using sleep labels from the PSG. Given that resting HR during wake is typically higher than during sleep, data from the last 30 minutes of collection that indicated wakefulness were discarded. The remaining HR data was then binned into 30-second intervals by median within each non-overlapping

window. Principal component analysis (PCA) was applied to normalized triaxial motion accelerometer data (x, y and z-directions) for improved feature extraction and noise reduction. To prevent data leakage, the training PCA transformation was applied to the test set, and motion magnitude was computed—after the 80-20 train-test split within subject. Outlier detection and removal were performed on a threshold of two standard deviations, and a simple moving average smoother (window size = 75 seconds) was used on the HR data to reduce artifact from Apple watch measurement.

C. Copula Fit and Prediction

A copula was fit using the TIl2nn methodology on the rank-normalized motion and HR data for each subject. The fit copula was then used to predict heart rate for the last 20% of the subject's data using only the triaxial accelerometer magnitude as input. The predicted heart rate was compared to the ground truth smoothed HR and the baseline prediction of mean HR for the training set. We note here that typical copula prediction functions support only bivariate inputs, where a joint CDF estimate for a new (x_1, x_2) is obtained using the empirical copula. Thus, the novelty of our methodology lies not only in the non-parametric copula estimate, but also in the prediction function that utilizes only motion-accelerometer data as input. Prediction in this study is performed by generating the empirical conditional cumulative distribution function (ECDF) at each rank-normalized motion point, which is then transformed back to the data space using the inverse marginal. Our novel addition to the methodology enables the copula to be used for multivariate prediction.

D. Model Evaluation

Spearman's rho (value between -1 and 1) was used to assess rank correlation between predicted and actual heart rate trajectories. Mean Directional Accuracy (MDA) (value between 0 and 1) evaluated whether the direction of the prediction aligned with the ground truth [8]. Spearman's rho and MDA were selected to quantify the model's ability to forecast overall trends in heart rate, rather than measuring accuracy as distance from ground truth. Finally, log-likelihood per observation was computed to gauge goodness-of-fit by standardizing over the total number of observations.

III. RESULTS

The results for all 26 subjects are presented in Table I, and Fig. 3 highlights the top three best- and worst-performing subjects alongside their corresponding contour plots. The selected cohort contained ~426 minutes of overnight HR and triaxial accelerometer data, with an average log-likelihood per observation of 0.36. With baseline Spearman's rho at 0 for constant mean prediction, the best-performing subjects 22, 10 and 2 exhibited statistically significant ($p < 0.05$) rho values ranging from 0.4 to 0.7. Most notably, subject 22 featured a rho of 0.738 with 0.509 MDA in comparison to baseline MDA of 0.081. Subject 22's heart rate time series did not display a stable cyclical trend but rather followed a convex trajectory, with heart rate increasing throughout the night before declining during the final third of recording. The other

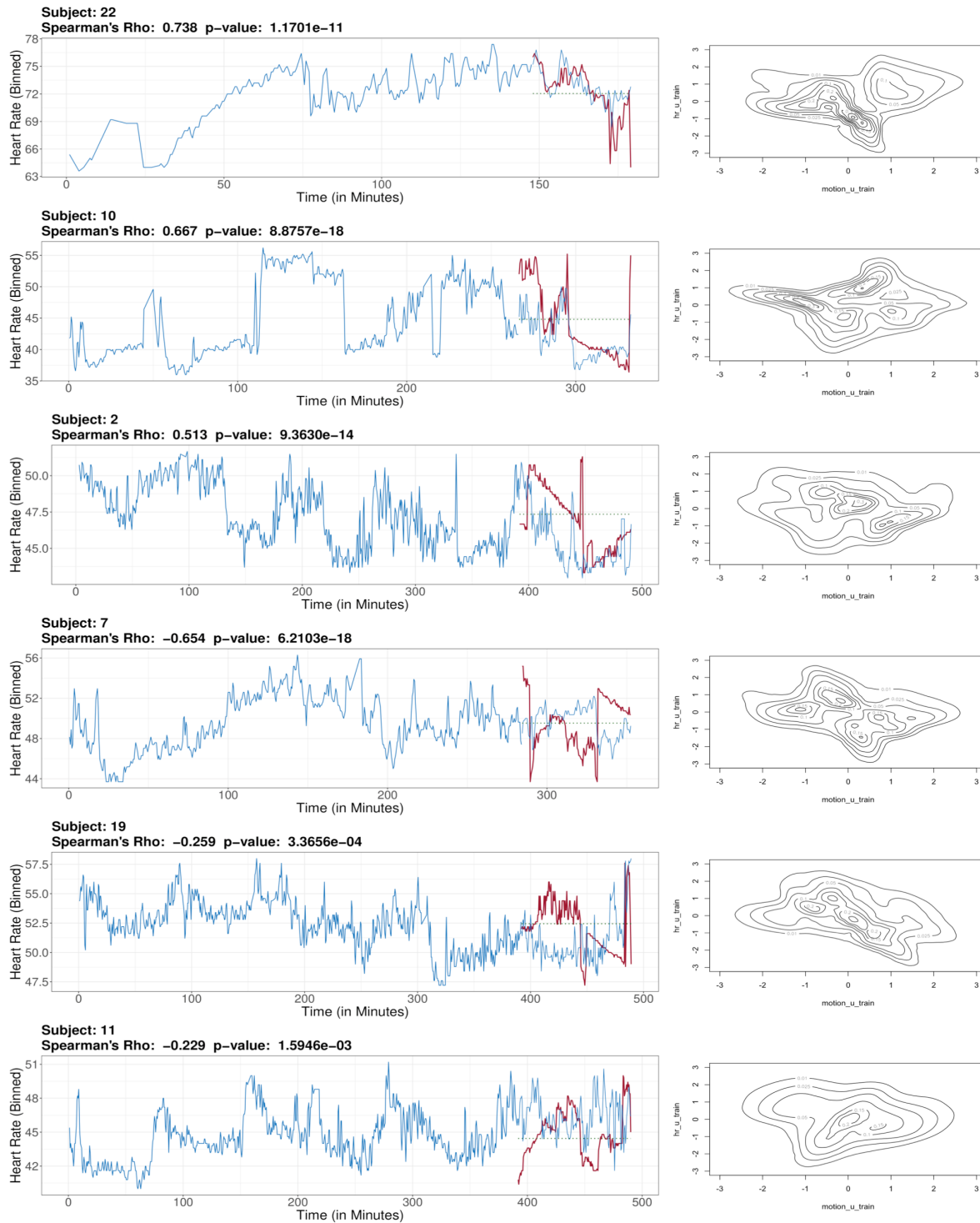


Fig. 3 Predicted and ground truth HR with empirical contour plots for the 3-best and 3-worst performing subjects. Spearman's rho and p-values featured.

two top-performing subjects (10 and 2) had more structured and cyclic heart rate patterns, with MDAs of 0.515 and 0.391 (baseline MDA 0.11 and 0.043 respectively), awarding accurate detection of HR peaks. Contour plots for top subjects further demonstrate a moderate-to-strong positive association between motion and heart rate. This finding aligns with physiological expectations since increased movement typically elevates heart rate due to greater metabolic demand. The three worst-performing subjects (out of the 4 performing below baseline) exhibited statistically significant

rho values between -0.7 and -0.3. In two of these cases (subjects 19 and 11), the model was trained on relatively consistent HR data. Both prediction algorithms effectively captured some peaks but struggled to map resting segments. Their MDA values 0.25-0.3 (for baseline MDA 0.13-0.17) highlight the same. The empirical copula contour plots for the worst-performing subjects revealed a dense, downward association that suggests the copulas learned the inverse physiological relationship—decreasing HR with increase

TABLE I. SUMMARY OF RESULTS BY SUBJECT

Subject ID	LLpO ¹	Rho ²	bMDA ³	MDA
1 (800685)	0.312	-0.013*	0.079	0.313
2 (6220552)	0.423	0.537	0.043	0.391
3 (8686948)	0.313	0.248	0.064	0.331
4 (1818471)	0.428	0.323	0.069	0.272
5 (3997827)	0.246	0.270	0.153	0.255
6 (1360686)	0.278	0.426	0.059	0.383
7 (5498603)	0.395	-0.654	0.169	0.296
8 (2598705)	0.361	0.205	0.136	0.299
9 (2638030)	0.387	0.264	0.148	0.367
10 (9961348)	0.715	0.667	0.110	0.515
11 (4426783)	0.174	-0.229	0.131	0.356
12 (4314139)	0.682	0.267	0.169	0.345
13 (9618981)	0.188	0.135*	0.114	0.298
14 (759667)	0.484	0.246	0.189	0.361
15 (46343)	0.317	0.195	0.046	0.382
16 (9106476)	0.229	0.231	0.103	0.375
17 (8173033)	0.288	0.192	0.138	0.343
18 (1449548)	0.309	0.122*	0.106	0.414
19 (8692923)	0.326	-0.259	0.151	0.256
20 (5383425)	0.311	0.257	0.224	0.427
21 (1455390)	0.342	0.183	0.102	0.340
22 (3509524)	0.410	0.738	0.081	0.509
23 (8258170)	0.277	0.459	0.180	0.493
24 (4018081)	0.319	0.239	0.176	0.394
25 (8530312)	0.284	0.213	0.110	0.313
26 (5797046)	0.233	0.199	0.082	0.297

¹LLpO is log-likelihood per observation; ²Starred values were **not** statistically significant ($p < 0.05$); ³bMDA is baseline MDA.

motion activity. We suspect this is due to empirical rank-transform issues or noisy data.

Across 26 subjects, the mean rho for individuals performing above a constant mean HR baseline was ~ 0.33 , with 0.37 MDA (baseline MDA 0.11). In aggregation, log-likelihood scores ranged 0.2-0.75 (Fig. 4). While a larger sample size is required to establish a definitive trend, preliminary results indicate a slight positive correlation (Pearson's correlation coefficient ~ 0.1) between the log-likelihood scores and Spearman's rho values. Poorer-fit models were likely unable to effectively learn underlying trends and struggled with prediction within subject.

IV. DISCUSSION

In this study, we explored a nonparametric copula approach using the Transformation Log-Likelihood Nearest Neighbors (Tll2nn) method with quadratic polynomials to model the dependence structure between acceleration-derived features and HR during sleep. By leveraging the empirical copula framework, we predicted heart rate from normalized acceleration data. Our results indicate that 20/26 subjects outperformed the baseline model, achieving an average Spearman's rho of ~ 0.33 and 0.37 MDA. This demonstrates the feasibility of the copula-based model for effective HR prediction. The variation in Spearman's rho values, ranging from -0.7 to 0.7, along with the empirical contour plots, indicates that strong model performance is typically observed when the copula captures the expected physiological relationship, as reflected in an upward-leaning contour plot. Conversely, downward-facing contours, which capture an inverse physiological dependency in certain individuals, are

associated with lower performance. In this context, it is important to note that the copula provides a significant advantage in model explainability and interpretability—an aspect that most black-box machine learning models lack; the generated contour plots allow us to visualize how the copula has “learned” key dependencies. As a proof-of-concept, this study demonstrates the potential of nonparametric copula modeling for uncovering latent dependencies in physiological data, without requiring strong parametric assumptions or large training datasets. This makes it especially well-suited for early-stage modeling, small-sample studies, or scenarios where model transparency is critical. Ultimately, future work should explore adaptive modeling, such as individualized parameter tuning, for optimized inter-subject performance.

Although our study highlights the viability of a novel copula-based model for heart rate prediction, limitations should be acknowledged. First, the sample size ($n = 26$) is relatively small, restricting the generalizability of our findings. While the 80-20 train-test split within subject helps mitigate overfitting, the limited data still risks poor generalization to unseen cases. Expanding the dataset and incorporating multi-night recordings would enable a richer exploration of subject-specific physiological variations. Finally, while Spearman's rho and mean directional accuracy (MDA) provide useful rank-based measures of association, they do not fully capture temporal dependencies in physiological signals. Future work may consider time-series-specific metrics and model design.

In conclusion, our study demonstrates the efficacy of a nonparametric copula approach for modeling heart rate dynamics using accelerometer data. The results suggest that such methods hold promise for real-world applications in wearable health monitoring, particularly for individualized heart rate tracking in sleep and clinical settings.

V. REFERENCES

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